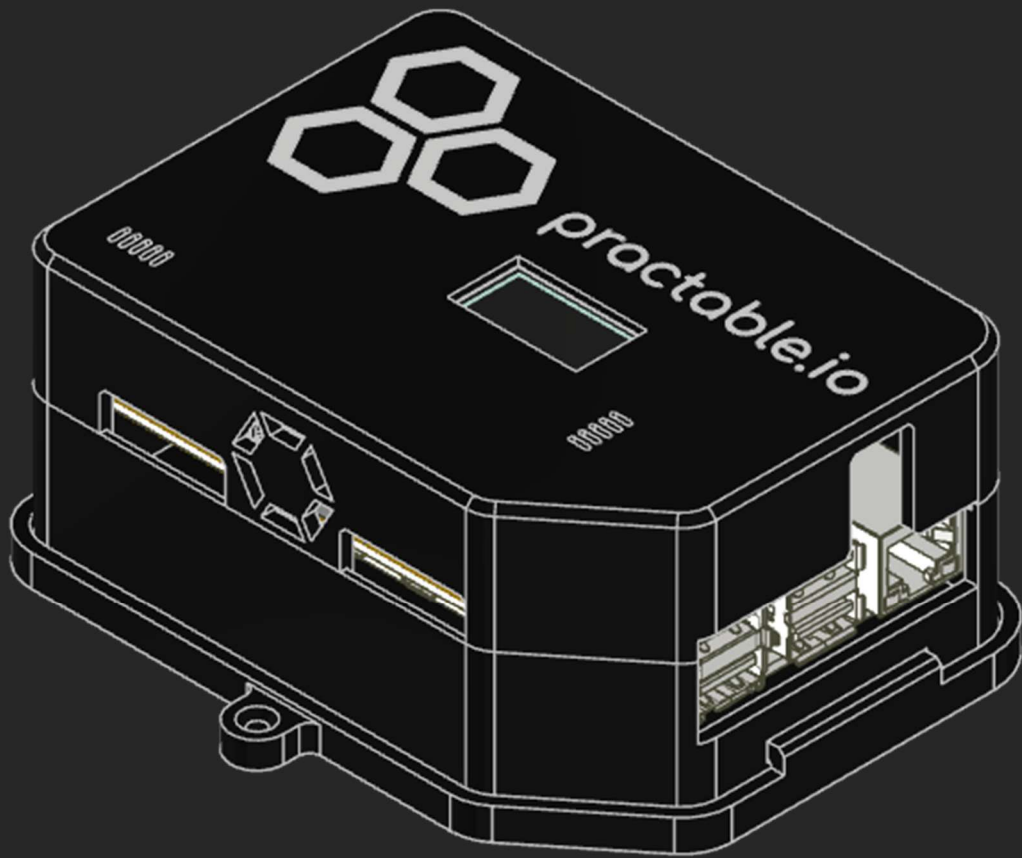




Practable: Portal

User Guide

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This equipment is supplied as prototype research hardware and is not classified as consumer electronic equipment.

The intended use case may require modification or integration of the hardware to meet specific customer or application requirements

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Author: Imogen Heard

Practable: Portal

Instant IoT Connectivity for Remote Lab Experiments

1. System Description

The Practable Portal is a turnkey solution for participating in the Practable.io Remote Labs project. It is pre-programmed to stream a webcam feed and relay USB serial communication between a connected microcontroller and the Practable.io servers, reducing the complexity of deploying experiments as remote labs.

The unit includes an adjustable power supply designed to support higher-current loads such as stepper motors and servos, providing a compact and integrated solution for deploying teaching experiments using microcontroller-based systems on the online platform.

1.1. Features

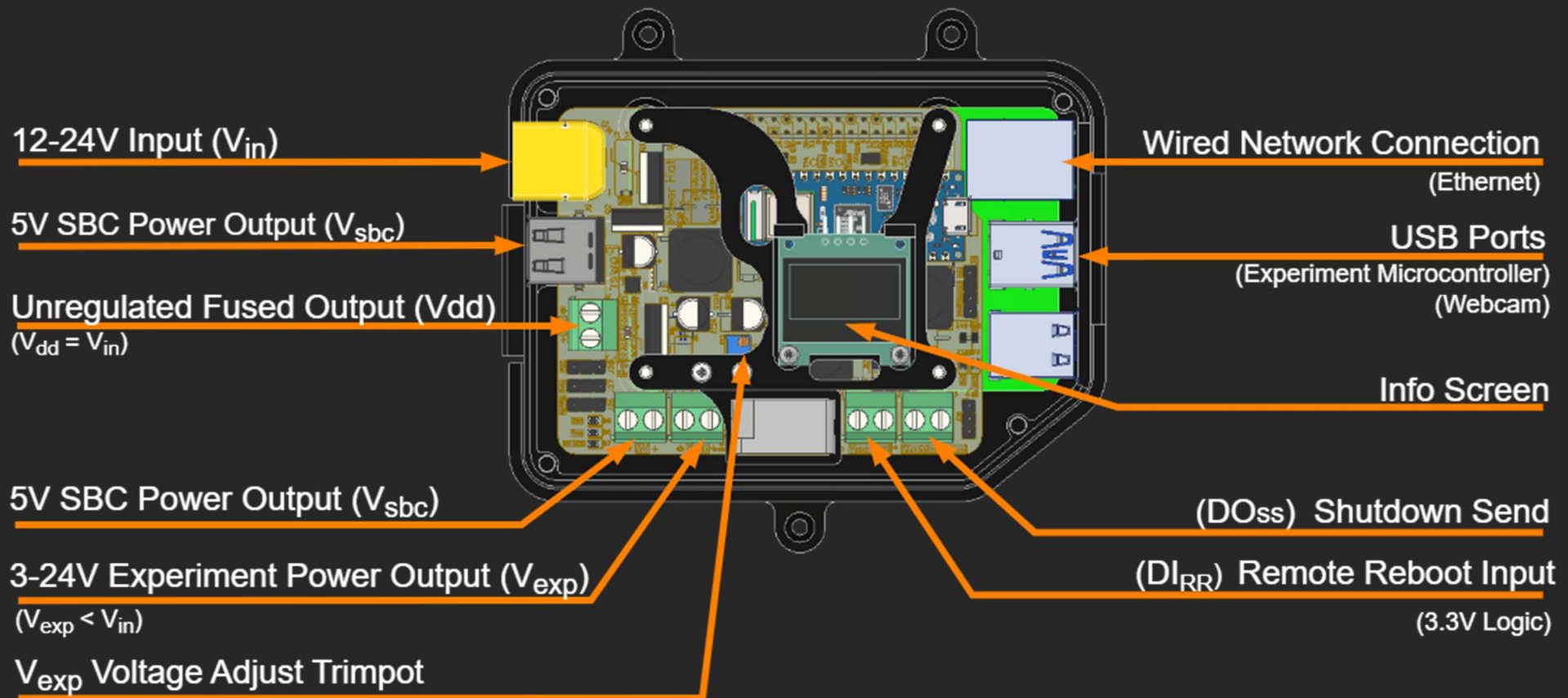
- 24 V input with polarity protection and fuse protection
- 5 V output for microcontrollers and logic-level devices (up to 2.5 A)
- User-adjustable output from 3.3 V to 24 V (up to 3 A)
- Integrated power supply for Raspberry Pi single-board computer (SBC)
- Voltage and current monitoring
- Display showing active network connection status
- Remote power cycle and reset functionality

1.2. System Components

- Raspberry Pi single-board computer (SBC)
- Pre-configured Raspberry Pi microSD card
- Practable Portal Raspberry Pi HAT and power supply
- Practable Portal enclosure and mounting plate
- 24 V DC power supply

2. System Specification

2.1. Hardware Overview



2.2. Electrical Specification

IO	Symbol	Condition	Min	Typ.	Max	Units	Comments
Supply							
Supply Voltage	V_{in}		6	24	24	V	
Supply Current	I_{in}	@ $V_{in} = 24\text{ V}$	0.190		3.5	A	Fused @ 3.5 A
Power Outputs							
Unregulated V_{out}	V_{dd}			V_{in}		V	
SBC Power V_{out}	V_{sbc}	$I_{max} = 3.0\text{ A}$	4.9	5.0	5.1	V	Power to SBC
Experiment Power V_{out}	V_{exp}	$I_{max} = 3.0\text{ A}$	3.3	12	24	V	Set via Trimpot
Digital IOs							
Shutdown Send	DOss		0		3.3	V	Send shutdown/reboot signal to remote
Remote Reboot Input	DlRR		0		3.3	V	Receive remote shutdown/reboot signal from remote

2.1. Hardware Description

2.1.1. Power Supply

The Practable Portal provides independent power outputs to help power your projects.

2.1.2. Adjustable Power Output (V_{exp})

This high-current output can supply up to 3 A and is adjustable from 3.3 V up to the input supply voltage, V_{in} (nominally 24 V).

The output voltage is adjusted using the Voltage Adjust Trimpot and a small flat-head screwdriver. Refer to [Section 3.3.2](#) for the voltage adjustment procedure.

2.1.3. 5V SBC Power Output (V_{sbc})

This fixed-voltage output powers the single-board computer (SBC) that runs the Portal software. It may also be used to power other external devices if required.

The output can supply up to 3 A total of which the SBC may draw up to 2.5 A.

2.2. Info Screen

The Info screen provides basic system information to assist with experiment setup, network monitoring, and troubleshooting potential system faults.

Three main screens are displayed in a continuous cycle while the equipment is powered on.

Power Stats Screen	Network Stats Screen	SBC Stats Screen
Practable Portal Experiment: 0.40 W 12.00 V 0.03 A SBC: 0.40 W 5.00 V 0.03 A	Practable Portal Runtime: 3:54:43 Last Ping: 0.0 IP: 10.42.0.21 MAC: 27:E6:AD:8B:A0:C6	Practable Portal Temp: 42.00 degC CPU: 0.10 % Mem: 8.70 % Disk: 24.26 %

2.2.1. Power Stats Screen

Displays status information for each power rail, including the adjustable experiment supply V_{exp} and the fixed 5 V SBC supply V_{sbc} for the Portal computer.

The displayed values are intended for basic power monitoring and fault finding only and should not be used for experimental data acquisition purposes.

2.2.2. Network Stats Screen

Displays network status information for the SBC, including System Runtime, Assigned IP and unit MAC addresses, and time since last ping to indicate valid internet connection.

An IP address will be displayed when the Practable Portal has established a valid network connection. This indicates that the Portal is connected to the local network and is able to communicate with Practable servers, provided a working internet connection is available.

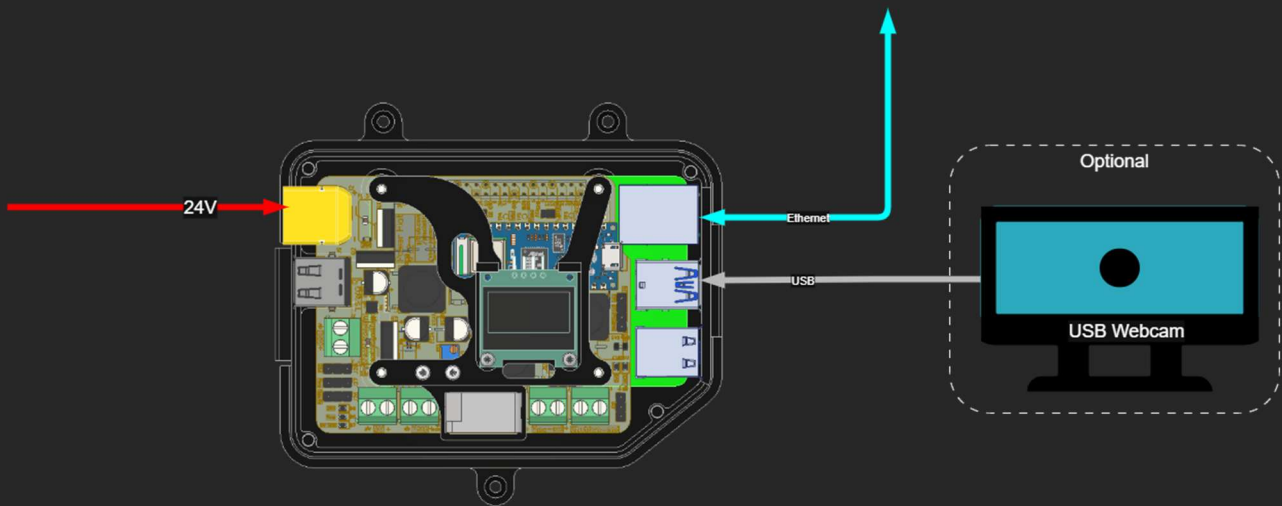
2.2.3. SBC Stats Screen

Displays additional SBC status information, including processor temperature, CPU, memory, and disk utilisation statistics.

3. Basic Setup

Practable Portal can be online in just a few simple steps.

3.1. Initial Setup



3.1.1. Network Connection

Connect an Ethernet cable from a LAN port to the Ethernet port on the Practable Portal. Within institutional or managed networks, permission from the network administrator may be required before connecting third-party devices to the wired network.

If the network administrator requires the MAC address of the Portal unit, power the device as described below and refer to the *Network Stats* page on the Info screen.

3.1.2. Webcam Connection (Optional)

If required, connect a USB webcam to the Practable Portal.

3.1.3. Power Connection

Connect the yellow XT60 connector on the Practable Portal chassis to the DC supply voltage. The input supply may range from 6 V to 24 V, with a nominal operating voltage of 24 V.

The input supply voltage determines the maximum available voltage of the adjustable experiment power output, V_{exp} .

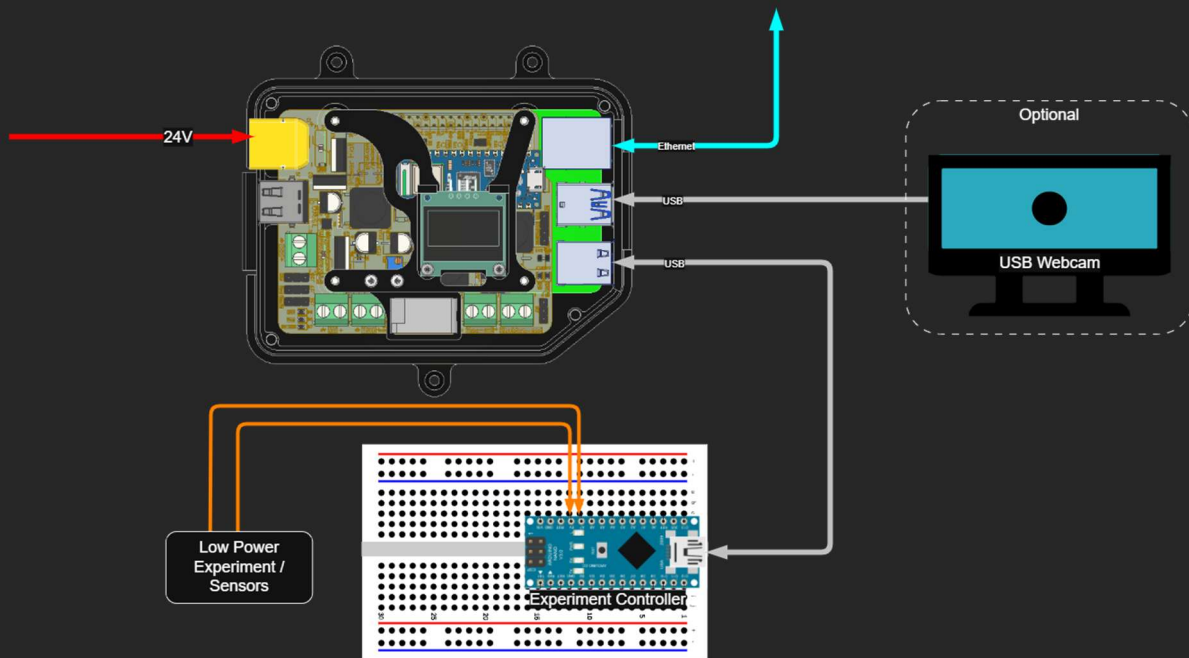
3.2. Checking Operation

If the Practable Portal has obtained a valid IP address through the LAN connection, this will be displayed on the *Network Stats* page of the Info screen.

If a USB webcam has been connected, it should now be active and streaming video, ready to be accessed through the Practable.io web service.

3.3. Setting up an Experiment

Setting up an experiment with the Practable Portal can be as simple as connecting a microcontroller to one of the USB ports on the right-hand side of the unit.

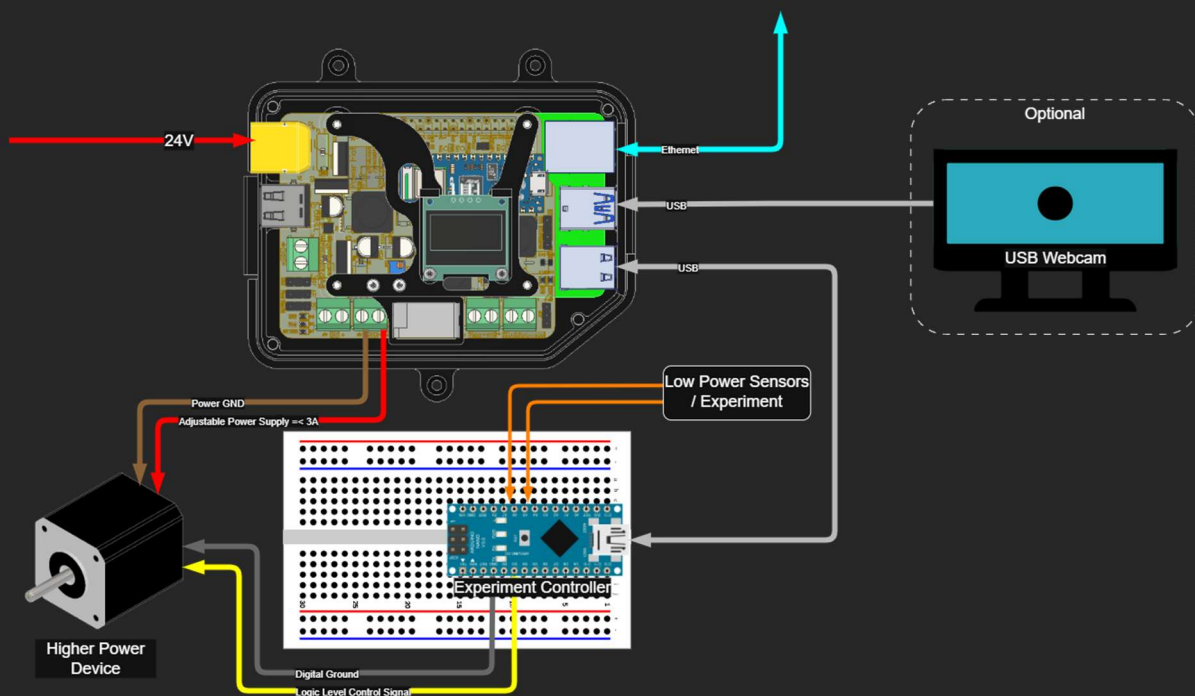


If the Practable Portal is already powered when a microcontroller is connected, the unit must be power-cycled before it can begin streaming and accepting data over USB Serial.

3.3.1. Using the Power Supply V_{exp}

For experiments requiring higher voltage or current than can be supplied via USB power alone, an additional power supply is available via the screw terminals on the Practable Portal.

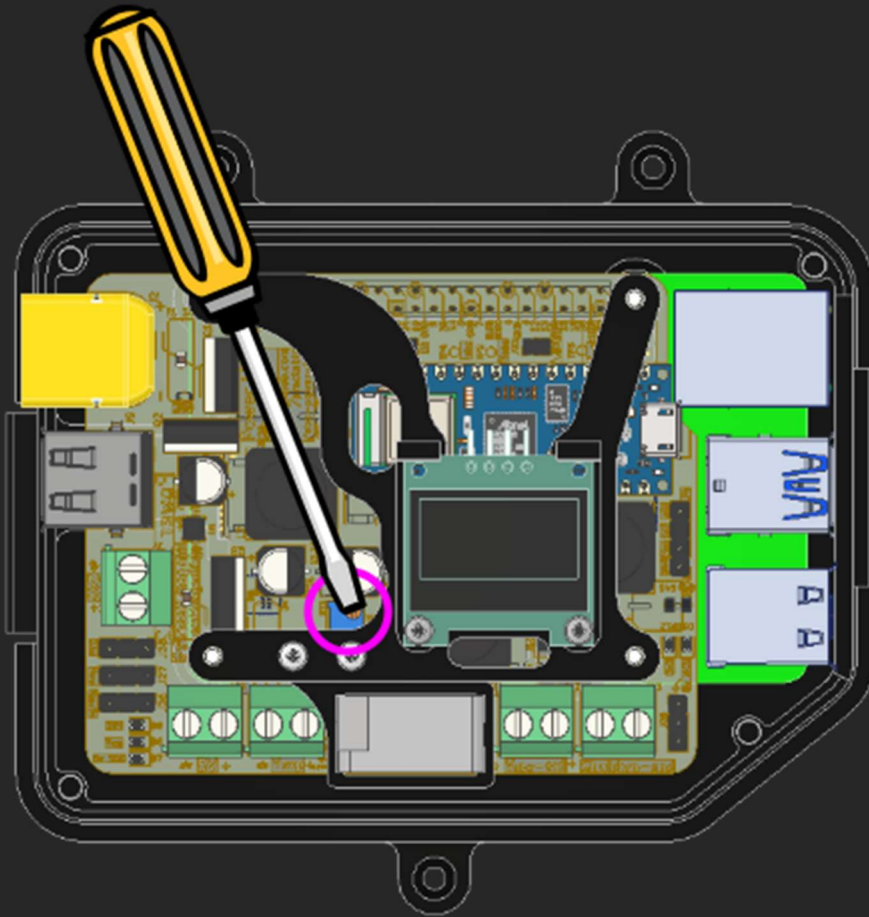
This power supply should be kept electrically separate from the microcontroller's digital ground. This helps ensure that high return currents from devices such as motors do not interfere with microcontroller operation.



3.3.2. Adjusting the Output Voltage V_{exp}

The voltage supplied by the Experiment Power Rail (V_{exp}) can be adjusted using a flat-blade screwdriver to turn the *Voltage Adjust Trimpot* (pink circle).

The currently measured output voltage is shown on the display. Adjust the trimpot slowly to ensure the displayed value remains accurate and stable.



If precise voltage setting is required, use an external multimeter to confirm correct voltage has been achieved.

4. Fault Finding & Debugging

4.1. No Network Connection

If Practable Portal does not have a valid network connection, it will not be assigned an IP address. This will be indicated by a blue box in the *IP* field of the *Network Stats* screen. It will likely be accompanied by *Last Ping* showing any number greater than 0.0

Network Stats Screen

Practable Portal	
Runtime:	3:54:43
Last Ping:	712
IP:	
MAC:	27:E6:AD:8B:A0:C6

4.1.1. Remediation

Please contact your Network Administrator and ensure that the device is allowed to connect to the LAN, and that it is placed on any whitelists required to enable external internet access.

4.2. Network Connection, No Internet

Practable Portal may have a valid LAN connection, and have been assigned an IP address, but is still unable to connect to the internet. In this case, the IP address will be displayed, but *Last Ping* will show a number greater than 0.0 and will be highlighted.

Network Stats Screen

Practable Portal	
Runtime:	3:54:43
Last Ping:	945
IP:	10.42.0.21
MAC:	27:E6:AD:8B:A0:C6

4.2.1. Remediation

- Please contact your Network Administrator and ensure that the device is placed on any whitelists required to enable external internet access.
- Check for valid internet access on other devices connected to the same network.
- Attempt to connect device using shared internet from another online device

4.3. Missing Data on Info Screen

If the power supply is unable to communicate with the Portal SBC for any reason, the Network Stats and SBC Stats screen will not populate with data, as shown below

Network Stats Screen

Practable Portal	
Runtime:	
Last Ping:	
IP:	
MAC:	

SBC Stats Screen

Practable Portal	
Temp:	degC
CPU:	%
Mem:	%
Disk:	%

4.3.1. Remediation

If system is otherwise functional, this may not require immediate fixing, as Practable Portal can continue to function in this state, however the following steps can be attempted to try and remedy issue.

- Power Cycle Unit
- If you have access to the device CLI run:
 - `systemctl status practable-portal.service`
 - Ensure "enabled" is returned
- Contact Practable.io if situation persists

5. Remote Rebooting

Although Practable Portals have proven to be highly reliable, with hundreds deployed across campus operating continuously, single-board computers are complex devices and may occasionally require a power cycle to restore normal operation.

If the device is locally accessible, this can be done by disconnecting and reconnecting the power supply. For installations where the device is located in a hard-to-reach area, the Practable Portal also supports remote power cycling, allowing it to power cycle itself and, where applicable, other connected devices.

5.1. Power Cycling Self

To power cycle a Practable Power unit, use the command line interface logged in as user *practable*.

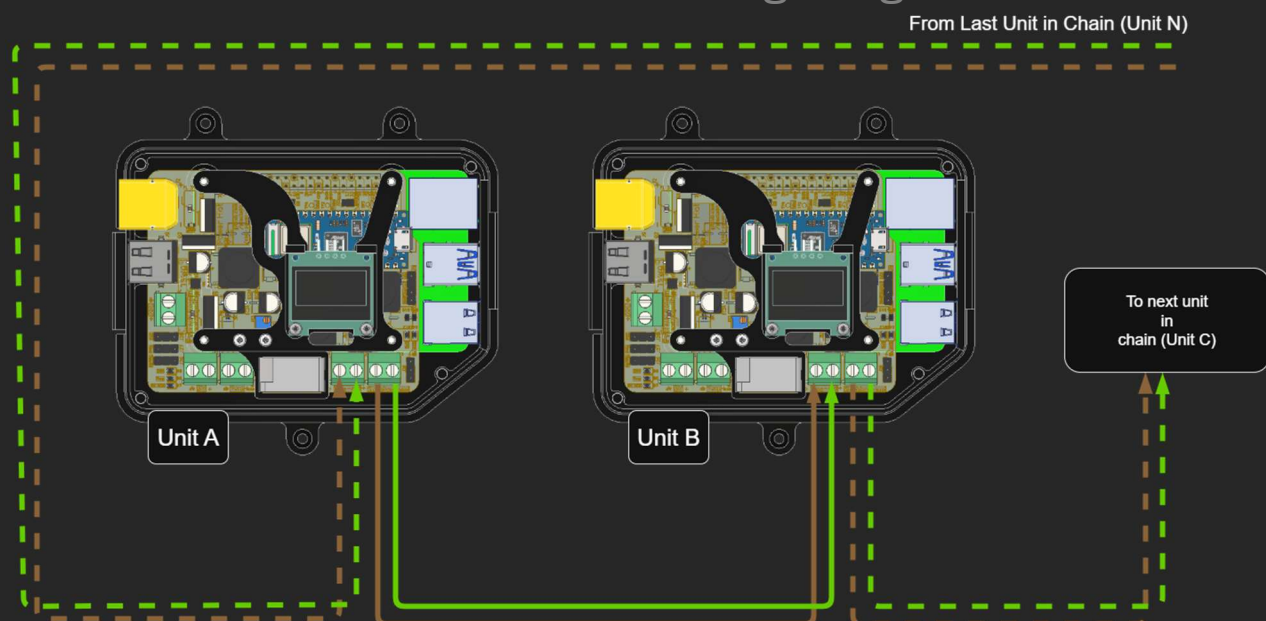
1. Run command
 - a. `./practable-portal/safe_power_cycle.sh`
 - b. Enter password for user *practable*
2. After ~3 seconds, the *Info Screen* will display *RX SBC Shutdown*. Power cycling procedure will commence after a pause to enable the system to safely shut down.

5.2. Power Cycling Neighbor Units

Power cycling neighbor units requires a two-wire interface (V+, GND) between each unit, from *shutdown-send* to *V_{sbc-ext}*.

We suggest wiring each unit to its neighbor in turn, and then wiring the last in the chain to the first in the chain. This way if a fault develops with any intermediate Portal that prevents access to the CLI, it can be externally power cycled by the unit before it in the chain. See Diagram below

5.2.1. Recommended Wiring Diagram



5.2.2. Sending Shutdown Command

To power cycle a Practable Portal (*Unit B*) via its neighbor (*Unit A*), log into the command line interface of *Unit A* as user *practable*.

1. Run command
 - a. `./practable-portal/send_shutdown.sh`
 - b. Enter password for user *practable*
2. After ~3 seconds, screen of *Unit B* will display *RX SBC Shutdown*. Power cycling procedure will commence after a pause.